

24.2 The Birth of Supersymmetry

If theoretical physics followed logic in its evolution, then after the proof of the Coleman–Mandula theorem someone seeking exceptions to this theorem would have noticed that it deals only with transformations that take bosons into bosons and fermions into fermions and are therefore generated by operators that satisfy commutation relations rather than anticommutation relations. This would have raised the question of whether a relativistic theory can have symmetries acting non-trivially on particle spins that take fermions and bosons into each other, and that therefore satisfy anticommutation relations rather than commutation relations. Exploring the possible structure of such a superalgebra along the lines described in the following chapter, supersymmetry would have emerged as the only possibility.

This is not what happened. Instead, supersymmetry arose in a series of articles on string theory and independently in a pair of little-noticed papers, about which more later, none of which show any sign that the authors were at all concerned with the Coleman–Mandula theorem.

Starting in the late 1960s, the effort to construct S -matrix elements for strong-interaction processes that would satisfy various theoretical requirements led to a new picture of various types of hadrons, as different modes of vibration of a string [6]. A point on a string labelled by a parameter σ will at time τ on some fixed clock have spacetime coordinates $X^\mu(\sigma, \tau)$, so the theory of a string’s motion in d spacetime dimensions may be regarded as a two-dimensional field theory with d bosonic fields, with action

$$\begin{aligned} I[X] &= \frac{T}{2} \int d\sigma \int d\tau \eta_{\mu\nu} \left[\frac{\partial X^\mu}{\partial \tau} \frac{\partial X^\nu}{\partial \tau} - \frac{\partial X^\mu}{\partial \sigma} \frac{\partial X^\nu}{\partial \sigma} \right] \\ &= T \int d\sigma^+ \int d\sigma^- \eta_{\mu\nu} \frac{\partial X^\mu}{\partial \sigma^+} \frac{\partial X^\nu}{\partial \sigma^-}, \end{aligned} \quad (24.2.1)$$

where T is a constant known as the string tension; $\mu = 0, 1, \dots, d-1$; and $\sigma^\pm$ are the two-dimensional ‘light-cone’ coordinates $\sigma^\pm \equiv \tau \pm \sigma$. This action can be derived from a more general version, with complete invariance¹ under transformations of a pair of ‘worldsheet coordinates’ σ_k

$$I[X] = -\frac{T}{2} \int d^2\sigma \eta_{\mu\nu} \sqrt{\text{Det } g} g^{kl} \frac{\partial X^\mu}{\partial \sigma^k} \frac{\partial X^\nu}{\partial \sigma^l}, \quad (24.2.2)$$

by passing to a special coordinate system in which the worldsheet metric g_{kl} satisfies the condition

$$\sqrt{\text{Det } g} g^{kl} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (24.2.3)$$

In much the same way that in electrodynamics the problems introduced by the negative sign of the action for timelike photons are eliminated by the gauge invariance of the theory, here the problems introduced by the negative sign of $\eta_{\mu\nu}$ in Eqs. (24.2.1) and (24.2.2) for $\mu = \nu = 0$ are eliminated by the invariance of the action (24.2.2) (for appropriate boundary

¹This symmetry is violated by quantum anomalies, like those discussed in Chapter 22, except in $d = 26$ spacetime dimensions for the purely bosonic theory, or $d = 10$ dimensions after the introduction of fermions.

conditions) under general transformations of the worldsheet coordinates. In the special coordinate system in which the action takes the form (24.2.1), there is an important remnant of invariance under general worldsheet coordinate transformations: invariance under the global *conformal transformations*:

$$\sigma^\pm \longrightarrow f^\pm(\sigma^\pm), \quad (24.2.4)$$

with $f^\pm$ a pair of independent arbitrary functions.

The particles described by this string theory do not match those seen in the real world. Ramond [7] and Neveu and Schwarz [8] in 1971, aiming respectively to introduce particles with half-integer spin or with the quantum numbers of pions, suggested the addition of d fermionic field doublets $\psi_1^\mu(\sigma, \tau)$ and $\psi_2^\mu(\sigma, \tau)$. Shortly after, Gervais and Sakita [9] introduced an action for this theory:

$$I[X, \psi] = \int d\sigma^+ \int d\sigma^- \left[T \frac{\partial X^\mu}{\partial \sigma^+} \frac{\partial X_\mu}{\partial \sigma^-} + i\psi_2^\mu \frac{\partial}{\partial \sigma^+} \psi_{2\mu} + i\psi_1^\mu \frac{\partial}{\partial \sigma^-} \psi_{1\mu} \right], \quad (24.2.5)$$

and noted that conformal invariance could be maintained by extending the conformal transformations (24.2.4) to act also on the fermion fields

$$\psi_1^\mu \longrightarrow \left(\frac{df^+}{d\sigma^+} \right)^{-1/2} \psi_1^\mu, \quad \psi_2^\mu \longrightarrow \left(\frac{df^-}{d\sigma^-} \right)^{-1/2} \psi_2^\mu. \quad (24.2.6)$$

Gervais and Sakita pointed out that, in addition to two-dimensional conformal invariance and d -dimensional Lorentz invariance, for appropriate boundary conditions this theory has a symmetry under infinitesimal transformations that interchange the bosonic field X^μ with the fermionic fields ψ_r^μ :

$$\begin{aligned} \delta\psi_2^\mu(\sigma^+, \sigma^-) &= iT\alpha_2(\sigma^-) \frac{\partial}{\partial \sigma^-} X^\mu(\sigma^+, \sigma^-), \\ \delta\psi_1^\mu(\sigma^+, \sigma^-) &= iT\alpha_1(\sigma^+) \frac{\partial}{\partial \sigma^+} X^\mu(\sigma^+, \sigma^-), \\ \delta X^\mu(\sigma^+, \sigma^-) &= \alpha_2(\sigma^-) \psi_2^\mu(\sigma^+, \sigma^-) + \alpha_1(\sigma^+) \psi_1^\mu(\sigma^+, \sigma^-), \end{aligned} \quad (24.2.7)$$

where α_1 and α_2 are a pair of infinitesimal *fermionic* functions of σ^+ and σ^- , respectively, like the Grassmann variables introduced in Section 9.5. This was an example of what has subsequently come to be called supersymmetry, a symmetry connecting bosons and fermions, but thus far it was only a symmetry of a two-dimensional field theory, not of a physical theory in four spacetime dimensions.

A few years later Wess and Zumino [10] referred back to the example of supersymmetry that had been provided by References [7], [8], and [9], and commented that it would be natural to try to extend the idea of supersymmetry to quantum field theories in *four* spacetime dimensions. They constructed several supersymmetric models. The simplest involved a single Majorana (self-charge-conjugate Dirac) field ψ , a pair of real scalar and pseudoscalar bosonic fields A and B , and a pair of real scalar and pseudoscalar bosonic

auxiliary fields F and G , with invariance under the infinitesimal transformation²

$$\begin{aligned}\delta A &= (\bar{\alpha}\psi), & \delta B &= -i(\bar{\alpha}\gamma_5\psi), \\ \delta\psi &= \partial_\mu(A - i\gamma_5 B)\gamma^\mu\alpha + (F - i\gamma_5 G)\alpha, \\ \delta F &= (\bar{\alpha}\gamma^\mu\partial_\mu\psi), & \delta G &= -i(\bar{\alpha}\gamma_5\gamma^\mu\partial_\mu\psi),\end{aligned}\tag{24.2.8}$$

where α is an arbitrary constant infinitesimal Majorana fermion c-number parameter. If we require invariance of the action under these transformations, then the most general real, Lorentz-invariant, parity-conserving, renormalizable Lagrangian density built out of these ingredients is

$$\begin{aligned}\mathcal{L} = & -\frac{1}{2}\partial_\mu A\partial^\mu A - \frac{1}{2}\partial_\mu B\partial^\mu B - \frac{1}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi \\ & + \frac{1}{2}(F^2 + G^2) + m\left[FA + GB - \frac{1}{2}\bar{\psi}\psi\right] \\ & + g\left[F(A^2 + B^2) + 2GAB - \bar{\psi}(A + i\gamma_5 B)\psi\right].\end{aligned}\tag{24.2.9}$$

Since the auxiliary fields F and G enter quadratically, we can derive an equivalent Lagrangian by setting them equal to the values given by the field equations

$$F = -mA - g(A^2 + B^2), \quad G = -mB - 2gAB.\tag{24.2.10}$$

The Lagrangian density then becomes

$$\begin{aligned}\mathcal{L} = & -\frac{1}{2}\partial_\mu A\partial^\mu A - \frac{1}{2}\partial_\mu B\partial^\mu B - \frac{1}{2}\bar{\psi}\gamma^\mu\partial_\mu\psi \\ & - \frac{1}{2}m^2[A^2 + B^2] - \frac{1}{2}m\bar{\psi}\psi \\ & - gmA(A^2 + B^2) - \frac{1}{2}g^2(A^2 + B^2)^2 - g\bar{\psi}(A + i\gamma_5 B)\psi.\end{aligned}\tag{24.2.11}$$

This Lagrangian density exhibits relations not only between scalar and fermion masses, but also between Yukawa interactions and scalar self-couplings, which are characteristic of supersymmetric theories. Wess and Zumino also described supersymmetry transformations and gave a Lagrangian for a supermultiplet containing a vector field. (We shall go into all this in more detail in Chapter 26.) Finally, in a second paper, Wess and Zumino [11] recalled the Coleman–Mandula theorem and traced the apparent violation of this theorem to the fact that the symmetry generators here satisfy anticommutation rather than commutation relations. It was a few more years before Gliozzi, Scherk and Olive [11a] showed that it was possible to construct a superstring theory with spacetime as well as worldsheet supersymmetry by imposing suitable periodicity conditions on the fields of the Ramond–Neveu–Schwarz model.

Unknown to Wess and Zumino, at the time of their first papers on supersymmetry in four spacetime dimensions this symmetry had already appeared in a pair of papers

²The notation for Dirac matrices used here is explained in the Preface and in Section 5.4. The γ_5 used here (which satisfies $\gamma_5^2 = \mathbf{1}$) is a factor of i times that used by Wess and Zumino, and the covariant conjugate $\bar{\psi}$ of any spinor ψ is defined here as i times that of Wess and Zumino. For this reason, some of the phases in Eqs. (24.2.8)–(24.2.10) are different from those in Reference [10].

published in the Soviet Union. In 1971 Gol'fand and Likhtman [12] had extended the algebra of the Poincaré group discussed in Section 2.4 to a superalgebra and used the requirement of invariance under this superalgebra to construct supersymmetric field theories in four spacetime dimensions. Their paper though prophetic gave few details, and was generally ignored until much later. Independently Volkov and Akulov [13] in 1973 discovered what today would be called spontaneously broken supersymmetry, but they used their formalism to identify the Goldstone fermion associated with supersymmetry breaking with the neutrino, an idea that met with no success. For most theorists, especially outside the Soviet Union, supersymmetry as a possible symmetry of nature in four spacetime dimensions began with the 1974 papers of Wess and Zumino.